

NOT Gate

Truth table

NOT Gate	
a	y
0	1
1	0

Behavioral source code (not_gate.v)

```
module not_gate(y,a);  
output y;  
input a;  
assign y=~a;  
endmodule
```

Testbench source code (not_gate_tb.v)

```
`timescale 1ns/1ps  
  
module not_gate_tb;  
// 1. Creating a register for the input and a wire for the output  
    reg test_a;  
    wire test_y;  
  
// 2. Instantiating the actual NOT gate module (connect pins)  
    not_gate uut(  
        .y(test_y),  
        .a(test_a)  
    );  
endmodule
```

```
// 3. Generate the input signals over time
initial begin
    // Telling iverilog to dump waveform data for GTKWave
    $dumpfile("simulation.vcd");
    $dumpvars(0, not_gate_tb);

    // Test case 1: a=0
    test_a = 0; #15;
    // Test case 2: a=1
    test_a = 1; #15;

    $finish; // End simulation
end
endmodule
```

fgfg

```
call C:\oss-cad-suite\environment.bat
```

fgfg

```
iverilog -o simulation.vvp not_gate.v not_gate_tb.v
```

fgfg

```
vvp simulation.vvp
```

Generate & check output waveform - Fig. 1(a)

```
gtkwave simulation.vcd
```

```
yosys
```

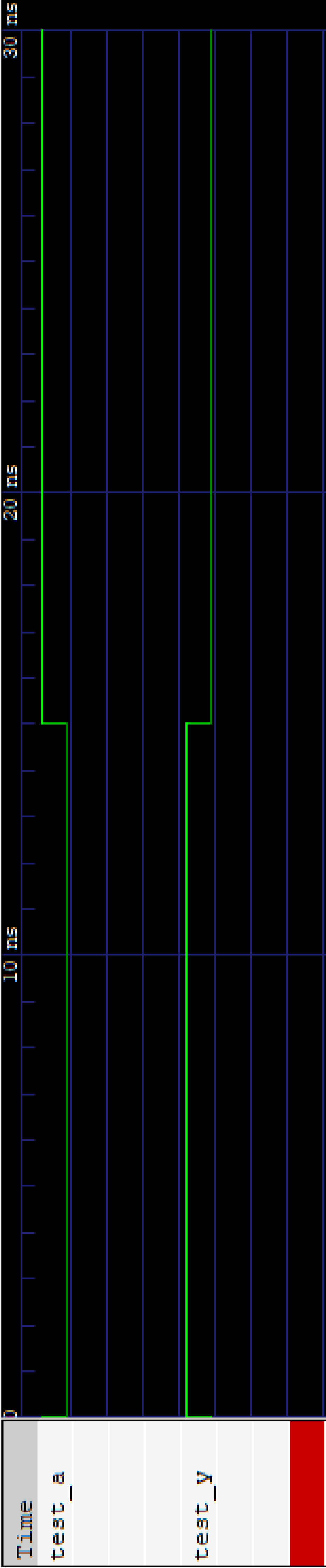


Fig. 3(a) - NOT gate output waveform

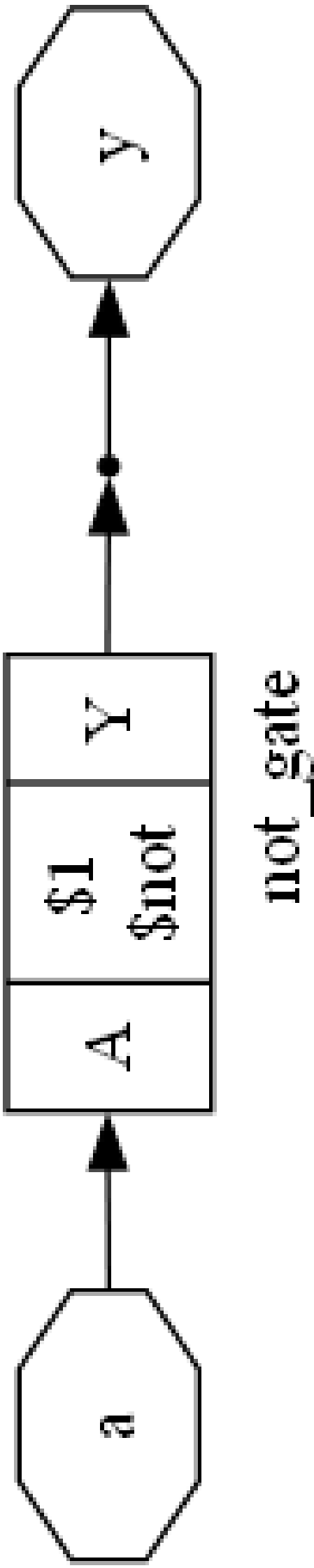


Fig. 3(b) - NOT gate generated schematic

```
/-----\
| yosys -- Yosys Open SYnthesis Suite                      |
| Copyright (C) 2012 - 2026  Claire Xenia Wolf <claire@yosyshq.com> |
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\-----/
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
```

```
yosys> read_verilog not_gate.v
```

```
yosys> hierarchy -top not_gate
```

```
yosys> proc opt techmap opt
```

```
yosys> write_verilog not_gate_netlist.v
```

```
yosys> show
```

```
5. Generating Graphviz representation of design.
Writing dot description to 'show.dot'.
Dumping module  not_gate  to page 1.
```

```
yosys> exit
```

```
Warnings: 2 unique messages, 3 total
End of script. Logfile hash: a730de9a18
Yosys 0.68+40 (git sha1 0f2bcb94b-dirty, Release, GNU /usr/bin/x86_64-w64-mingw32-g++ 15.2.1)
Time spent: 5% 2x write_verilog (0 sec), 5% 2x read_verilog (0 sec), ...
```

```
[OSS CAD Suite] D:\[redacted]\NOT>"D:\[redacted]\windows_10_cmake_Release_Graphviz-15.1.1-win64\Graphviz-15.1.1-win64\bin\dot.exe" -Tpng show.dot -o schematic.png
```

